

Detection of Faults in Electrical Panels Using Deep Learning Method

Dragan Mlakić

Distribution Company, Electric Power Company HZ-HB
Inc. Mostar, Vilenička bb 72295, Novi Travnik,
Bosnia and Herzegovina
dragan.mlakic@ephzhh.ba

Srete Nikolovski, Zoran Baus

Faculty of Electrical Engineering, Computer Science and
Information Technology Osijek
Josip Juraj Strossmayer University of Osijek, Osijek, Croatia
srete@ferit.hr, zoran.baus@ferit.hr

Abstract— *In the image analysis, a big trend within the field of artificial intelligence is using the Deep Learning method, which is an upgrade of the existing neural network adaptive architecture (ANN). Deep Learning is a major new field in machine learning that encompasses a wide range of neural network architectures designed to perform various tasks. In the thermography energy sector, examples that are processed on a daily basis are sampling of active energy components, focus segmentation, and fault classification. The most popular network architecture for Deep Learning in image analysis is the convolution neural network (CNN), where traditional machine learning methods require determination and calculation, from which the algorithm training comes. Deep Learning approach captures important features as well as the appropriate weight of these attributes to make decision for new data. This paper describes a method and tool that are available to build and conduct an effective analysis of the Deep Learning Method for electrical components.*

Keywords— *Deep Learning; electrical components; thermal imaging; picture analysis; Convolution Neural Network Introduction*

I. INTRODUCTION

Deep learning (DL) has been dramatically improved as a method for solving various tasks primarily for artificial intelligence (AI), such as object detection, speech recognition, translation of machine behavior, etc. The very essence of DL architecture is one that allows addressing many more complex problems that are primarily predicated for AI [1]. As a result, scientists have expanded the DL field of action to many of the problems that we encounter on a daily basis, such as: face, speech, and motion detection. Using reversal methods for neural networks (i.e. Back Propagation), DL is able to address the aforementioned situational problems in real time. Several researchers have used Neuronal Networks (NN) in their research to successfully model the method of conclusion for the above-mentioned problems [5-7]. Many previous articles are based on pure neural network application to recognize objects in images based on human visual cortex. However, there is no practical example of usage in situations like real time protection systems, online fault recognition, power reduction based on heat transfer.

This paper presents one situation where one of Deep Learning method is suitable for usage in electricity installations for fault recognition. Other researches are conducted in focus of harmonic detection with thermal

recording [9], oil aging of high voltage transformer with thermal analysis [10], and this paper presents new approach to IR image analysis with same goal, damage prevention.

For the purpose of this paper, only one of the above-mentioned architectural DL modifications is explained, which is specifically created for the recognition of the faces in the images, the Convolution Neural Network (CNN). The tool used is AlexNet, as part of the Matlab software package that enables the introduction of hand-built architectures into its user request framework. The first chapter deals with topics related to power components of energy network in the field of safety and efficiency of the low voltage distribution network equipment for the supply of equipment for automatic control of the component. This is an analysis of thermography images of the terminal block with emphasis on the intensity and distribution of heat. In the second chapter, CNN is processed as basis of the analysis discussed in the first chapter, and a package of learned rules adapted to the thermography picture of the terminal block is also provided. Using of the Matlab tool, the thermography image are uploaded and the results of the analysis are desired output. Chapter 3 provides a practical example of CNN application within Matlab over images from IR cameras.

II. ELECTROMAGNETIC COMPONENTS OF THERMAL ANALYSIS

Premature switch off equipment in the distribution network usually results in a loss of electricity production, increased management and maintenance costs, unsatisfied customers and ultimately loss of economic gain. Companies for primary engineering and insurance companies discover that after analysis of losses, about 30% of the costs fall into the failure of the electrical equipment, according to ASNT (The American Society for Nondestructive Testing) [8].

The failure diagnosis method suitable for power plants, without physical contact and the acceptable measurement results is the thermographic recording of equipment done with thermal camera (Infra-Red-IR). IR thermography is the only diagnostic technology that enables instant insight and confirmation of thermal capabilities by seeking anomalies and allowing the user to combine them with precision, without temperature measurement contact. As a measurement result, the user receives a real automatic document in the form of a

visual presentation of the recorded object and can easily include in the final report.

For the purpose of this paper, the content is limited of low-voltage distribution panels and control equipment, but the diagnosis method applies to all electrical components with regard to the physical properties of the material itself. In practice, before some electrical component burns, it warms up first. The heating of the electrical component, is the first sign that there is a malfunction. Some of the faults that are encountered when components are heating: weak connections, non-balanced loads, defective switches, damaged switching equipment, overcurrent circuits, poor quality of human operation on equipment installation, poor quality of material. Preventive identification of these failures is critical to the maintenance and protection of the electrical plant and end users of the system. Showcase mentioned in Fig 1 presents a bad galvanic connection between the cable lugs and the coil housings. The consequence of this is increased temperature on the coil due to energy losses that also make potential possibility of fire on the coil.

The consequence of this condition is the withdrawal of insulation on the cable and the dissolution of the PVC clamp components of the terminal clamp. An elevated temperature does not necessarily mean a malfunction of electrical equipment, but in combination with other parameters, it is an indicator of disturbance in the system. The volume of disturbance can be changed at certain moments, disrupting the relationship between physical electrical components, and reduce the impact of the variable component on the overall system security. Risk assessment and model of action can't be made based on one or more thermal images rather than constant monitoring of affected low voltage (LV) network components via thermal images.

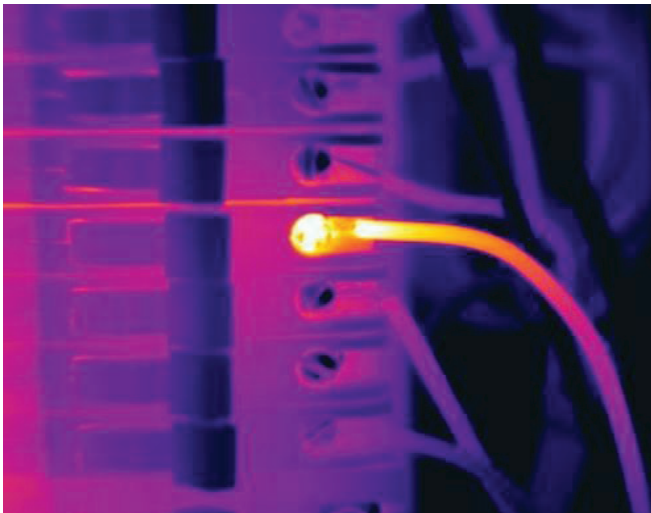


Fig. 1. Thermal image of the galvanic connection of the cable lugs and the housing of the distribution cabinet

The smallest electrical components can have a huge impact on the efficiency of the system we are analyzing. If not identified at the time, the elevated temperature may accumulate and cause the components to dissolve. Another

situation, presented in Fig. 2, is an Infra-Red capture of an AC/DC transformer located in the automatic control cabinet increased heating causes disturbances of power on PLC board and thus their operational condition. In addition, the risk of melting a PVC AC / DC transformer casing is large if the heat load continues.

Behind the heat irradiation stands energy exchange, that shows potential risk of failure. In order to prevent this negative process, it is necessary to set a focus on preventive maintenance. Well timed and quality information is essential for achieving that goal. By using IR recording with the help of Artificial Intelligence methods for preventive maintenance can be achieved in much higher degree.



Fig. 2. IR power supply of AC/DC power transformer cabinet for distribution control panel

III. CONVOLUTION NEURAL NETWORK

The Convolutional Neural Network (CNN) algorithms mainly develop from knowledge of the human visual cortex. The design of CNN is based on the biological architecture of the human visual system.

In order to understand CNN, it is necessary to understand the functionality of the human visual cortex. The visual cortex is located on the back of the human head, part of the cerebral cortex that plays an important role in processing visual information received in the brain [2], [5]. The visual information received through the eye goes through a series of brain structures and comes to a visual cortex. The visual cortex receives input from a sensor (eye) known as the primary visual cortex (V1). The information is further managed through the splitting field, including the V2 field of vision and the V4 field of vision. There are other visual fields (V3, V5 and V6), but we will not process them in this report. V1, V2, and V4 are part of the ventral visual system responsible for recognizing the shape, related to the presentation of complex parts of objects, such as face shape. Fig. 3 illustrates the visual cortex ventricular system.

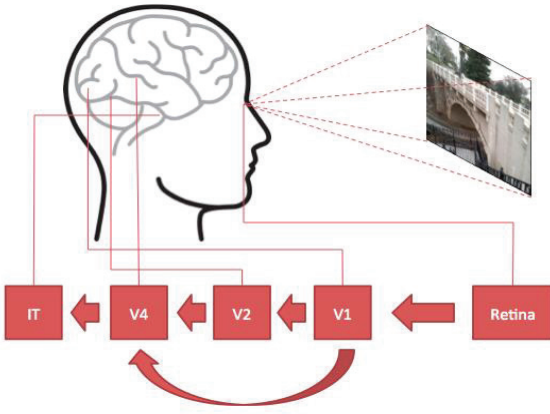
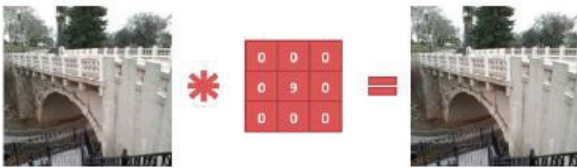


Fig.3. Ventral Visual Cortical System.

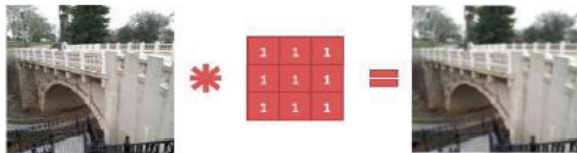
The retina transforms the light energy that is refractory from the objects to the chemical energy. This chemical energy is transformed into electrical impulses that are transmitted to the visual cortex.

V1 (primary visual cortex) usually does the task of recognizing the boundaries of objects where the boundary of a region is a strong contrast in the visual signal. V2 (secondary visual cortex) is the first part of the visual cortex associated with the association. It gets a strong forward link from V1 and jumps strong links to other parts of the visual cortex. V2 processes signals related to orientation, spatial frequency, color, and several other signals. V4 records object-related properties of medium complexity, such as simple geometric shapes, as an adjunct to orientation, spatial frequency, or color. Inferior time gyrus (TI) is responsible for recognizing color-based and shape-based objects compared to processed information in object memory. TI does semantic tasks, such as face recognition. The aforementioned parts of the visual cortex referred to CNN for a user exposed to technical terms. CNN is composed of the following components: the aggregate layer and under sample layer. CNN operations are strictly mathematical operations with two functions (f and g) that produce the third function (h), which is the integral amount of function f moved across another function g; the form of the formula is:

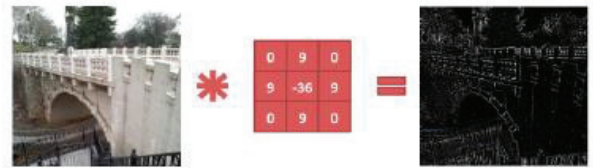
$$h(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau \quad (1)$$



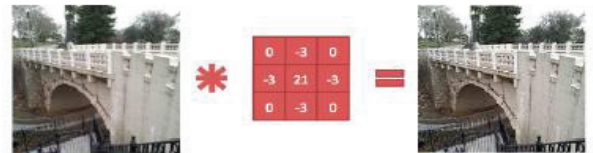
(a) Identification grain



(c) Blur grain



(b) Border grain



(d) Sharpe grain

Marked as $h=f * g$.

CNN usually works with 2-D group operations that may be represented as shown in Fig. 4.

22	15	1	3	60
42	5	38	39	7
28	9	4	66	79
0	2	25	12	17
9	14	2	51	3

 $*$

0	0	1
0	0	0
1	0	0

 $=$

29	12	64
38	41	32
13	80	81

Fig. 4. A simple illustration of a 2-D group operation

Left part is the input matrix, (presented in Fig. 4.); the middle is called the "grain" matrix. The aggregate operation is performed on these matrices and the matrix obtained on the right part of Fig. 4. The aggregate process is based on the elements followed by the sum of elements. When the upper left part of the input matrix (3x3) is "convoluted" with the grain matrix, the number 29 is obtained. Then we move one field to the right and re-segment 3x3 convoluluses with the grain matrix and we get number 12. In the same way, we start by the input matrix. We are doing a batch operation and write results in the output matrix. Since the matrix is 3x3, each target matrix must be 3x3. If we get 5x5, then we reduce it to 3x3 matrix after a batch operation. This means that any schedule to be in the 3x3 matrix in the 5x5 input arrays, the output number will be 29.

This location neglect of the schedule is of crucial importance for visual problems. Ideally, recognizing the shape of an object in an image should be undesirable to its rotation or orientation in the image or some of its parts. In CNN, this problem is elegantly solved by a batch metric operation. To link the visual cortex and aggregate operation, we must assume that the input matrix is of 3x3 images to be processed. Fig. 5 shows the analogy of visual cortex and aggregate surgery with matrices.

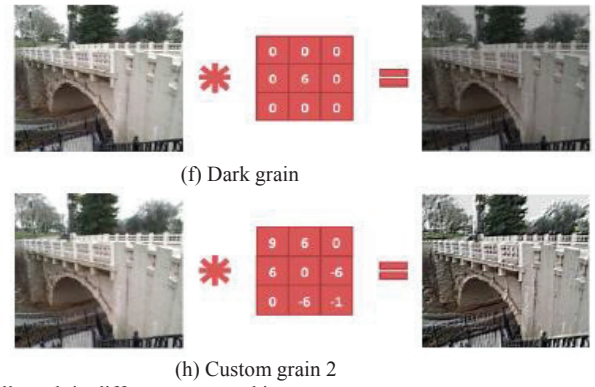
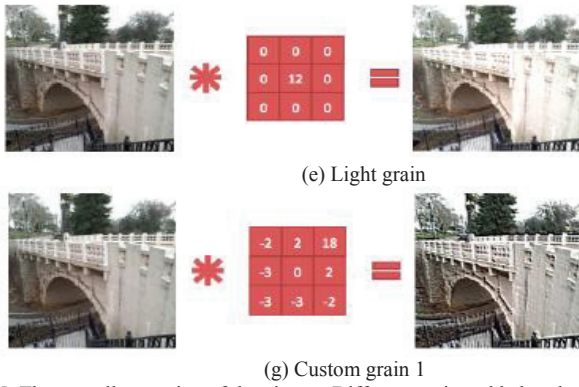


Fig. 5. The overall operation of the picture. Different grains added to the same image will result in different processed images

Various batches allow different analyzes to work on the image. It is possible and recommended to combine several grains in the same analysis. In Fig. 5, under (b) a grain that isolates the boundaries of the objects in the picture is depicted, and it is one of the roles of a visual cortex in the human brain, although, CNN has alternatives in form of RNN (Recurrent Neural Network) [11]. RNN is deep neural network architecture for object recognition based on modification of replacing the ubiquitous convolution in combination with pooling layer of the deep CNN with four RNN that sweep horizontally and vertically in both directions across the image. The tool for RNN is called ReNet, which is alternative to AlexNet, mentioned below.

III. CASE SAMPLE

Used case for DL goal locating fault is simulated on switchboard in low voltage distribution cabinet. The main goal is locating heat dissipation. DL algorithm is able to send alert or/and report to user about cabinet condition, depending on speed of heat dissipation, high spot temperature and location. In this way the greater damage could be avoided, in case that the heat is high enough to induce fire or losses in power. Besides that, possible damage could be avoided when the temperature is rising with the constant current. User needs to provide IR images of cabinet regardless of the temperature of the cabinet (fault or not fault). In this way, the difference of temperature will be easily spotted by DL. The other possible way is to provide images only without fault but, in that case, the user has to train it through CNN to report on any difference in images in future. All images are taken via thermal (IR) camera

In order for Matlab to be able to work with pictures provided by the user, it is necessary to introduce a trained CNN network that is able to distinguish the details in the images. This network can be downloaded as add-ons to Matlab via the Mathworks Web site or via the "Add-On" function in Matlab itself. In the CNN network, we are looking for "Alexnet", which is the result of training on several thousand different images to make the CNN network successful when an unknown object is entered as an image input. The part that is needed for this example is one feature located within one of

the image analysis channels that focuses on only one part of the image. We first download the CNN network and instantiate it in a different variable.

Command:
net = alexnet;

Load the image and show it on the screen. The image will be automatically stored in the Matlab workspace. Fig. 6 shows uploaded IR image in workspace ready for processing. Command:

```
im=imread(fullfile(matlabroot,'examples','nnet','IR_0215.jpg'));  
imshow(im)  
imgSize = size(im);  
imgSize = imgSize(1:2);
```

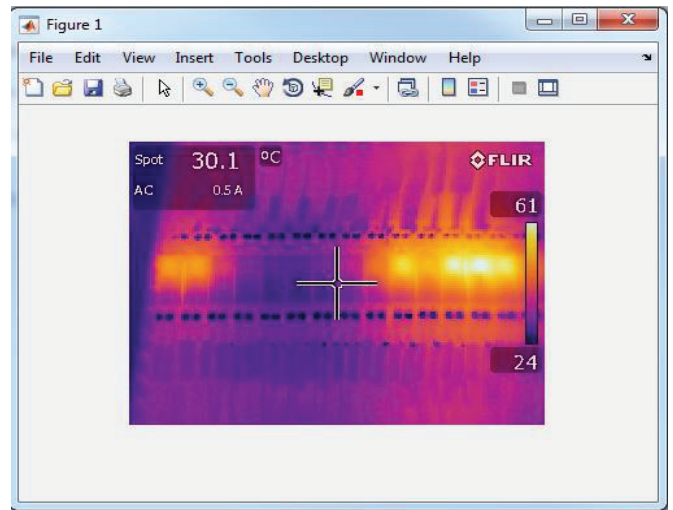


Fig. 6. Original IR camera image uploaded to the Alexnet network and stored in Matlab workspace

To see what layers we have at our disposal so that we can use it with our picture, we need listing of functions. The aggregate algorithm uses layers with the learned parameters. The network itself learns to use features, usually one feature per layer. Command:

net.Layers
ans =

25x1 Layer array with layers:

1	'data'	Image Input	227x227x3 images with 'zerocenter' normalization
2	'conv1'	Convolution	96 11x11x3 convolutions with stride [4 4] and padding [0 0]
3	'relu1'	ReLU	ReLU
4	'norm1'	Cross Channel Normalization	cross channel normalization with 5 channels per element
5	'pool1'	Max Pooling	3x3 max pooling with stride [2 2] and padding [0 0]
6	'conv2'	Convolution	256 5x5x48 convolutions with stride [1 1] and padding [2 2]
7	'relu2'	ReLU	ReLU
8	'norm2'	Cross Channel Normalization	cross channel normalization with 5 channels per element
9	'pool2'	Max Pooling	3x3 max pooling with stride [2 2] and padding [0 0]
10	'conv3'	Convolution	384 3x3x256 convolutions with stride [1 1] and padding [1 1]
11	'relu3'	ReLU	ReLU
12	'conv4'	Convolution	384 3x3x192 convolutions with stride [1 1] and padding [1 1]
13	'relu4'	ReLU	ReLU
14	'conv5'	Convolution	256 3x3x192 convolutions with stride [1 1] and padding [1 1]
15	'relu5'	ReLU	ReLU
16	'pool5'	Max Pooling	3x3 max pooling with stride [2 2] and padding [0 0]
17	'fc6'	Fully Connected	4096 fully connected layer
18	'relu6'	ReLU	ReLU
19	'drop6'	Dropout	50% dropout
20	'fc7'	Fully Connected	4096 fully connected layer
21	'relu7'	ReLU	ReLU
22	'drop7'	Dropout	50% dropout
23	'fc8'	Fully Connected	1000 fully connected layer
24	'prob'	Softmax	softmax
25	'output'	Classification Output	crossentropyx with 'tench', 'goldfish', and 998 other classes

The network can train with images of much larger dimensions, but it requires a lot of processor time and work memory. The next step is to research features by looking at areas in the aggregated layer activated in relation to the original image. Layers activated at heat-shrinking locations are searched and the focal points on the thermal block are located. Based on the number of focal points, their intensity concludes that the thermal block is in the mode of operation that is not normal and so we conclude a possible failure and raise the alarm. Each layer CNN consists of several 2D rows called "channels".

Release the image through the network and examine the output activation of the conv1 layer. Command:

```
act1 = activations (net,im,'conv1','OutputAs','channels');
```

Activations return as 3D rows, with a third dimension indexed on the "conv1" layer. To see these activations using the "montage" function, it's needed to change the order in 4D. The third dimension in the "montage" input represents the color of the image. Set the third dimension to 1, so the color will be omitted because the intensity of the light on the picture does not require color, as in Fig. 7. The fourth dimension is the channel index. Command:

```
sz = size(act1);
act1 = reshape(act1,[sz(1) sz(2) 1 sz(3)]);
montage(mat2gray(act1),'Size',[8 12])
```

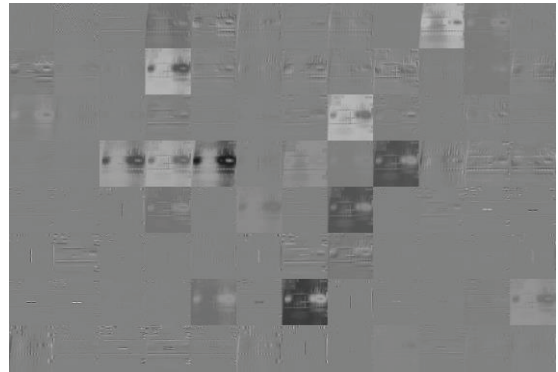


Fig. 7. Channel activation for image analysis without color and with light on individual images.

Every activation can have any value. This is why the output normalizes with the "mat2gray" function. All activations are scaled with minimum 0 and maximum 1. White pixels represent positive activation, and black pixels represent strong negative activation. The mostly gray channel does not activate so strongly in the original picture. The pixel position on the activation channel exists in correspondence with the same position on the original image. The white pixel on the same location on the channel shows strong activation of that channel on the original image. In Fig. 8. channel 36 has the same size as the original image. Command:

```
act1ch36 = act1(:,:,,36);
act1ch36 = mat2gray(act1ch36);
act1ch36 = imresize(act1ch36,imgSize);
imshowpair(im,act1ch36,'montage')
```

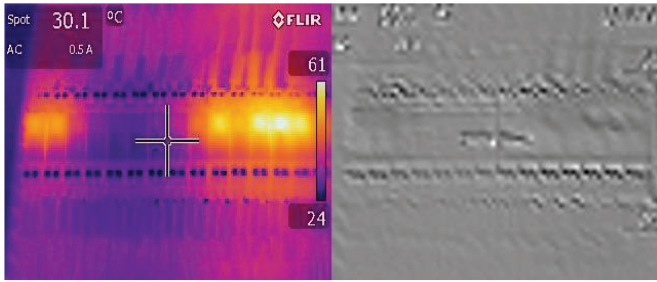


Fig. 8. Channel 36 in size as the original image and original image left.

This channel is activated in red because the white pixels are in the position on the original picture in the place of red color, reference in Fig. 9. It is possible to extract the channel with the strongest activation. Command:

```
[maxValue,maxValueIndex] = max(max(max(act1)));
act1chMax = act1(:, :, maxValueIndex);
act1chMax = mat2gray(act1chMax);
act1chMax = imresize(act1chMax, imgSize);
imshowpair(im, act1chMax, 'montage')
```

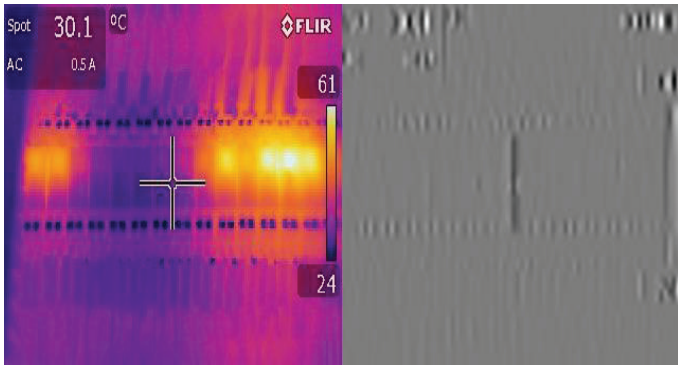


Fig. 9. The strongest activation channel. The channel changes to the black and white transition on the original image

CNN learns to notice features, such as color or shapes, in the image at its first bulk level. At deeper aggregate levels, the network learns to see details that are more complicated by combining the results from previous levels, presented in Fig. 10. For the purpose of this example, examine the "conv5" layer in the same way as "conv1". Command:

```
act5 = activations(net, im, 'conv5', 'OutputAs', 'channels');
sz = size(act5);
act5 = reshape(act5, [sz(1) sz(2) 1 sz(3)]);
montage(imresize(mat2gray(act5), [48 48]))
```

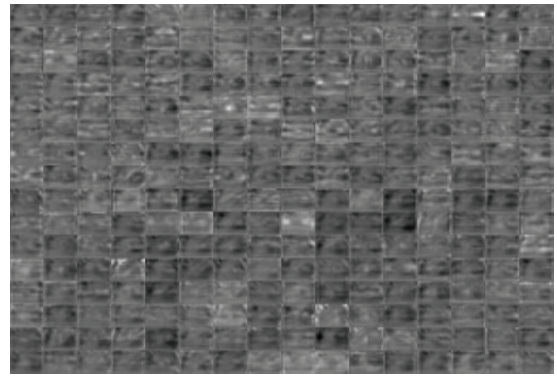


Fig. 10. Conv5 channel presents layers suitable for analysis

Focus on the most active layer. Command:

```
[maxValue5,maxValueIndex5] = max(max(max(act5)));
act5chMax = act5(:, :, maxValueIndex5);
imshow(imresize(mat2gray(act5chMax), imgSize))
```



Fig. 11. Most active channel on conv5 layer regarding original image

This channel focuses on the white color and shapes of the picture that are not perfect for our analysis. With a more detailed analysis of the presented channels, the number 36 focuses on parts of the original image, presented in Fig. 12. Command:

```
montage(imresize(mat2gray(act5(:, :, [36])), imgSize))
```

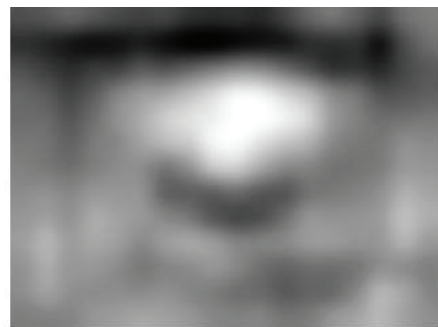


Fig. 12. Channel 36 with focus on the bright parts of the original image.

Repeat the analysis to visualize activation of the "relu5" layer. In order to leave only positive activations, the correction unit (ReLU) is used reference in Fig. 13. Command:

```
act5relu = activations (net,im,'relu5','OutputAs','channels');
sz = size(act5relu);
act5relu = reshape (act5relu,[sz(1) sz(2) 1 sz(3)]);
montage (imresize (mat2gray(act5relu(:,:,:[36])),imgSize))
```



Fig. 13. ReLu activated on conv5 layer and obvious focal point.

Compared to the activation of the "conv5" layer, activation on the "relu5" layer clearly shows the zones that are thermal centers on the original image. To notice a change in heat output, it is necessary to compare the current image with a new image from the same angle shot with the same camera at another time so that the difference in the pixels in the "relu5" layer is accurately visualized.

Command:

```
pic2 = imread (fullfile(matlabroot,'examples','nnet','face-eye-closed.jpg'));
imshow(imClosed)
pic2act = activations (net, pic2,'relu5','OutputAs','channels');
sz = size(act5Closed);
pic2act = reshape (act5Closed,[sz(1),sz(2),1,sz(3)]);
```

To compare the image command:

```
channelsPic2 = repmat (imresize(mat2gray(pic2act (:,:[3 5])),imgSize),[1 1 3]);
channelsOpen = repmat(imresize(mat2gray(act5relu(:,:,:[3 5])),imgSize),[1 1 3]);
montage(cat(4,im,channelsOpen*255, pic2, channelsPic2*255));
title('Input Image, Channel 3, Channel 5');
```

The presented script can be placed inside an infinite loop and saved as .m file, and thus run on the camera in real time. The script that is run will then work with the analysis of the previous image and the new received image, and give results on the change of focus on the terminal block, the intensity of the blow, the temperature and, if needed, the thermal energy that is emitted. Based on this information, it is possible to act in a timely manner in order to prevent failure in the distribution cabinet, potential fire.

IV. CONCLUSION

The paper presents the AI Deep Learning method with a CNN network trained and tested for recognition of various objects in the images. Proposed method is presented in one use case of low voltage wiring cabinet. Due to similarity of CNN with human visual cortex logic behind technical implementation of algorithm of CNN in tool, AlexNet is explained. The method used for the analysis of finding faults on the electro-energy components of the distribution network. From the report, in Chapter 3 it can be seen that the CNN network is very flexible and thus suitable for situations requiring analysis of experts from the area from which the situation is. Through the carefully trained NN network with carefully selected impulses, algorithms with operations and matrices that skillfully simulate human conclusion and thereby reduce the resources needed for the process of analysis. Ultimate result is timely action on protecting people and equipment. Further research on this method will compare results against other tools, like above-mentioned ReNet.

REFERENCES

- [1] B. Erickson, P. Korfiatis, Z. Akkus, and K. Philbrick, "Toolkits and libraries for deep learning," Journal of Digital Imaging, March 2017. DOI: 10.1007/s10278-017-9965-6
- [2] G.E. Hinton, S. Osindero, and Y-W. Teh, "A fast learning algorithm for deep belief nets," Neural Computation, MIT Press, 2006.
- [3] A. Krizhevsky, I. Sutskever, and G.E. Hinton, "Imagenet classification with deep convolutional neural networks," Advances in Neural Information Processing Systems, vol. 25, 2012.
- [4] H. Wang and B. Raj, "On the origin of deep learning," February 2017.
- [5] K. Osako, R. Singh, and B. Raj, "Complex recurrent neural networks for denoising speech signals," in Proc. of the IEEE Workshop on Applications of Signal Processing to Audio and Acoustics, 2015. DOI: 10.1109/WASPAA.2015.7336896
- [6] A. Gupta, H. Wang, and M. Ganapathiraju, "Learning structure in gene expression data using deep architectures, with an application to gene clustering," in Proc. of the IEEE International Conference on Bioinformatics and Biomedicine, November 2015. DOI: 10.1109/BIBM.2015.7359871
- [7] L.A. Gatys, A.S. Ecker, and M. Bethge, "A neural algorithm of artistic style," Journal of Vision, August 2015. DOI: 10.1167/16.12.326
- [8] J.H. Shazly, M.A. Mostafa, D.K. Ibrahim, E.E. Abo El Zahab, "Thermal analysis of HV cables with several types of insulation for different configurations in the presence of harmonics", IET Generation Transmission and Distribution, June 2017. DOI: 10.1049/iet-gtd.2016.0862
- [9] S. Harjo, R.A. Pasaribu, "Thermal aging of mineral oil - Paper composite insulation for high voltage transformer", International Journal on Electrical Engineering and Informatics, vol. 8, no. 4, pp. 820-835, December 2016. DOI: 10.15676/ijeei.2016.8.4.9
- [10] ANSI/ASNT, "ASNT Standard for qualification and certification of nondestructive testing personnel", CP-189-2011, no. 2509, 2016.
- [11] F. Visin, K. Kastner, K. Cho, M. Matteucci, A. Courville, and Y. Bengio, "ReNet: A recurrent neural network based alternative to convolutional networks," arXiv:1505.00393, May 2015.